NBA DATA

Kaden Garvin

2023-11-27

```
library(readr)
nbaData <- read_csv("Desktop/nbaData")
```

```
## Rows: 12846 Columns: 67
## ── Column specification ────────────────────────────────────────────────────────
## Delimiter: ","
## chr  (4): PLAYER_NAME, NICKNAME, TEAM_ABBREVIATION, season_id
## dbl (63): PLAYER_ID, TEAM_ID, AGE, GP, W, L, W_PCT, MIN, FGM, FGA, FG_PCT, F...
##
## ℹ Use `spec()` to retrieve the full column specification for this data.
## ℹ Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
library(tidyverse)
```

```
## Warning: package 'dplyr' was built under R version 4.2.3
```

```
## Warning: package 'stringr' was built under R version 4.2.3
```

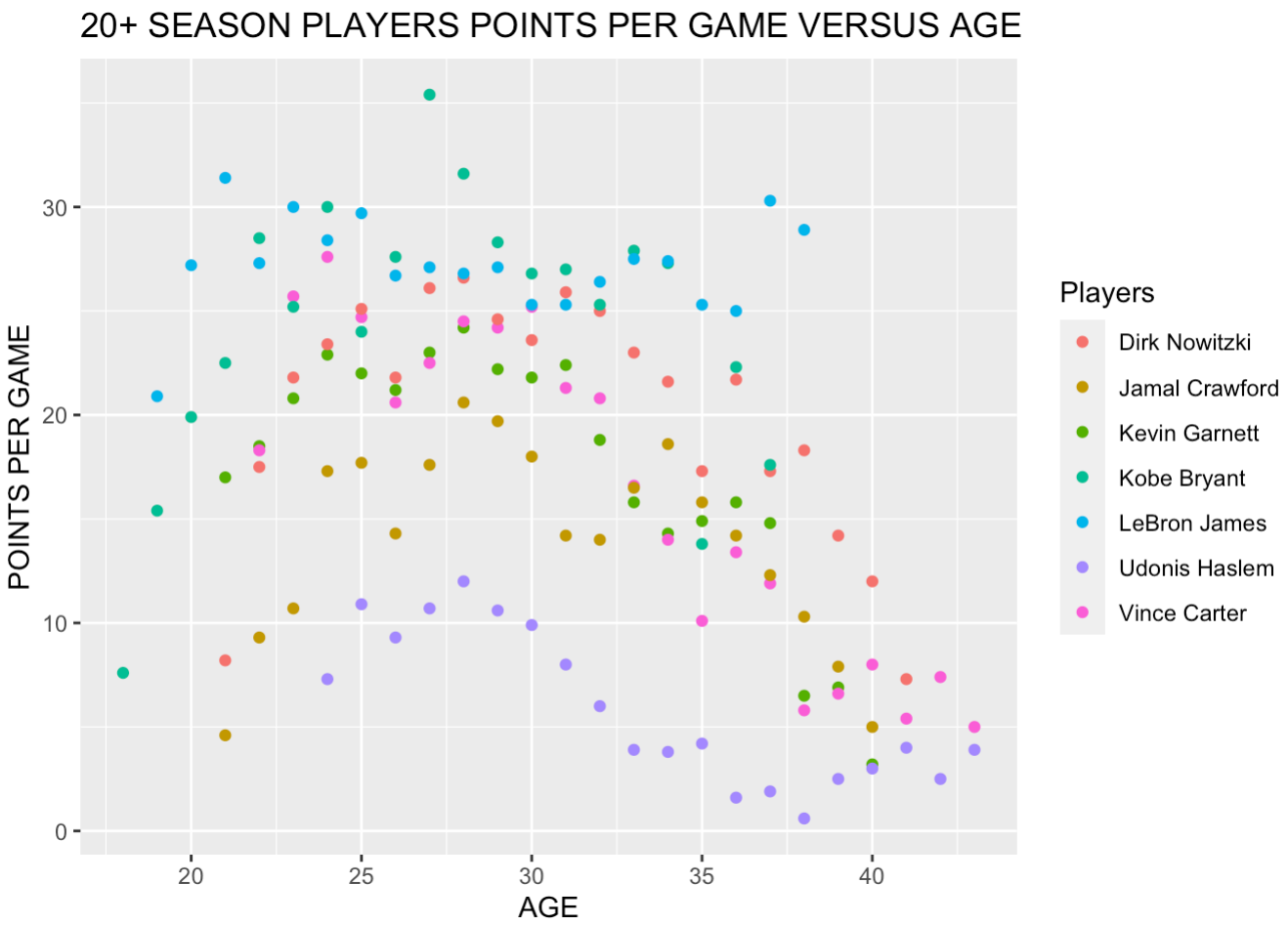
```
## ── Attaching core tidyverse packages ──────────────────────── tidyverse 2.0.0 ──
## ✔ dplyr     1.1.4     ✔ purrr     1.0.2
## ✔ forcats   1.0.0     ✔ stringr   1.5.1
## ✔ ggplot2   3.4.4     ✔ tibble    3.2.1
## ✔ lubridate 1.9.3     ✔ tidyr     1.3.0
## ── Conflicts ────────────────────────────────────────── tidyverse_conflicts() ──
## ✖ dplyr::filter() masks stats::filter()
## ✖ dplyr::lag()    masks stats::lag()
## ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
lebron <- nbaData %>%
  filter(PLAYER_NAME %in% c("LeBron James")) %>%
  select(PTS,AGE,GP,OREB,DREB,REB,AST,TOV,STL,BLK,season_id,MIN,NBA_FANTASY_PTS,W_PCT,NICKNAME)
```

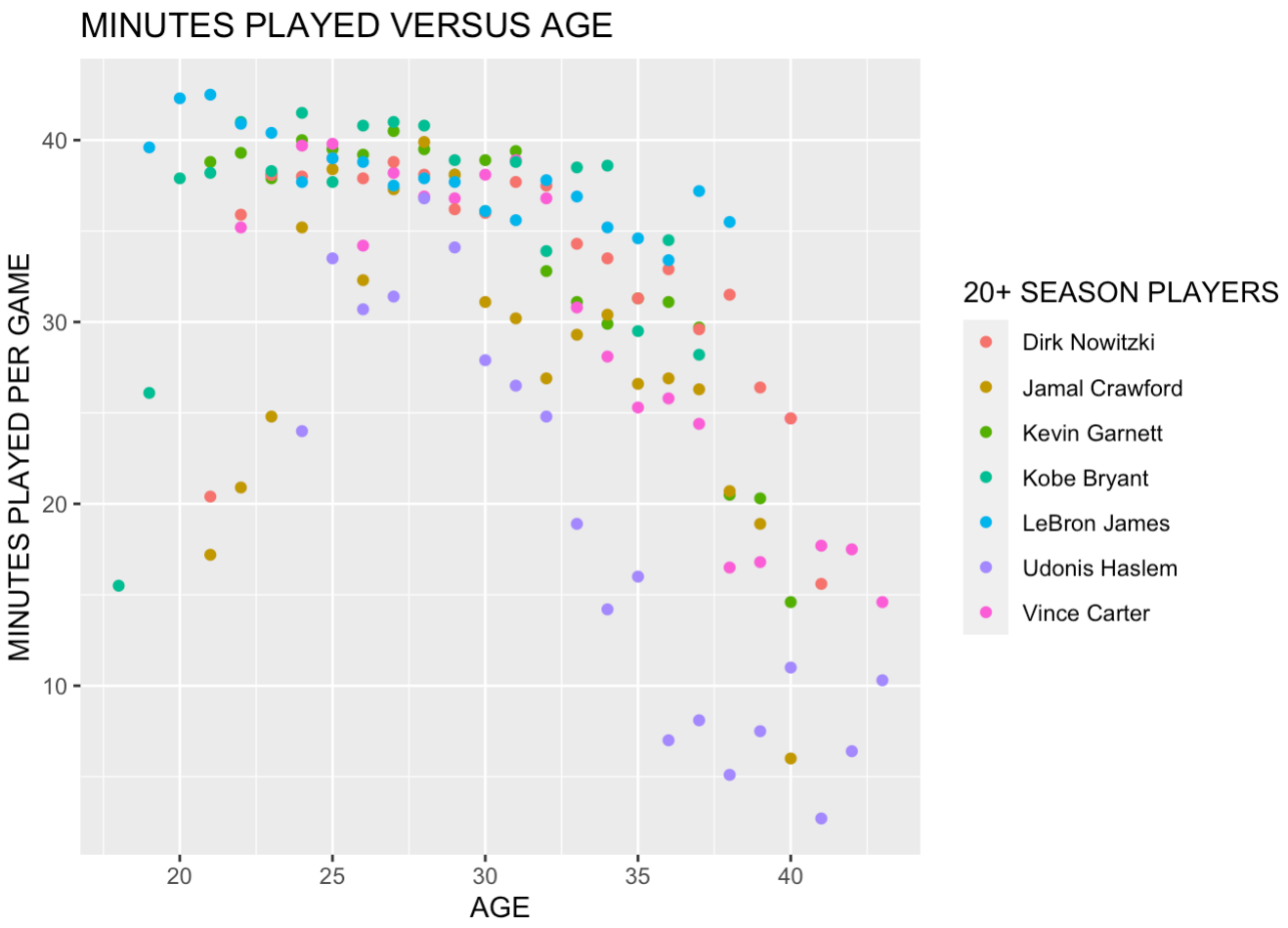
```
summary(lebron)
```

```
##       PTS             AGE              GP             OREB           DREB
##  Min.   :20.90   Min.   :19.00   Min.   :45.00   Min.   :0.60   Min.   :4.200
##  1st Qu.:26.12   1st Qu.:23.75   1st Qu.:65.75   1st Qu.:1.00   1st Qu.:6.000
##  Median :27.15   Median :28.50   Median :76.00   Median :1.15   Median :6.400
##  Mean   :27.20   Mean   :28.50   Mean   :71.05   Mean   :1.16   Mean   :6.400
##  3rd Qu.:28.52   3rd Qu.:33.25   3rd Qu.:79.00   3rd Qu.:1.30   3rd Qu.:7.025
##  Max.   :31.40   Max.   :38.00   Max.   :82.00   Max.   :1.80   Max.   :7.500
##       REB             AST              TOV             STL
##  Min.   :5.500   Min.   : 5.900   Min.   :3.000   Min.   :0.900
##  1st Qu.:7.225   1st Qu.: 6.525   1st Qu.:3.300   1st Qu.:1.300
##  Median :7.650   Median : 7.200   Median :3.450   Median :1.600
##  Mean   :7.540   Mean   : 7.340   Mean   :3.500   Mean   :1.515
##  3rd Qu.:8.050   3rd Qu.: 7.925   3rd Qu.:3.625   3rd Qu.:1.625
##  Max.   :8.600   Max.   :10.200   Max.   :4.200   Max.   :2.200
##       BLK         season_id              MIN        NBA_FANTASY_PTS
##  Min.   :0.300   Length:20          Min.   :33.40   Min.   :40.00
##  1st Qu.:0.600   Class :character   1st Qu.:35.98   1st Qu.:47.95
##  Median :0.700   Mode  :character   Median :37.70   Median :51.30
##  Mean   :0.745                      Mean   :37.83   Mean   :50.55
##  3rd Qu.:0.900                      3rd Qu.:39.15   3rd Qu.:53.15
##  Max.   :1.100                      Max.   :42.50   Max.   :55.80
##      W_PCT          NICKNAME
##  Min.   :0.4180   Length:20
##  1st Qu.:0.5825   Class :character
##  Median :0.6710   Mode  :character
##  Mean   :0.6466
##  3rd Qu.:0.7288
##  Max.   :0.8150
```

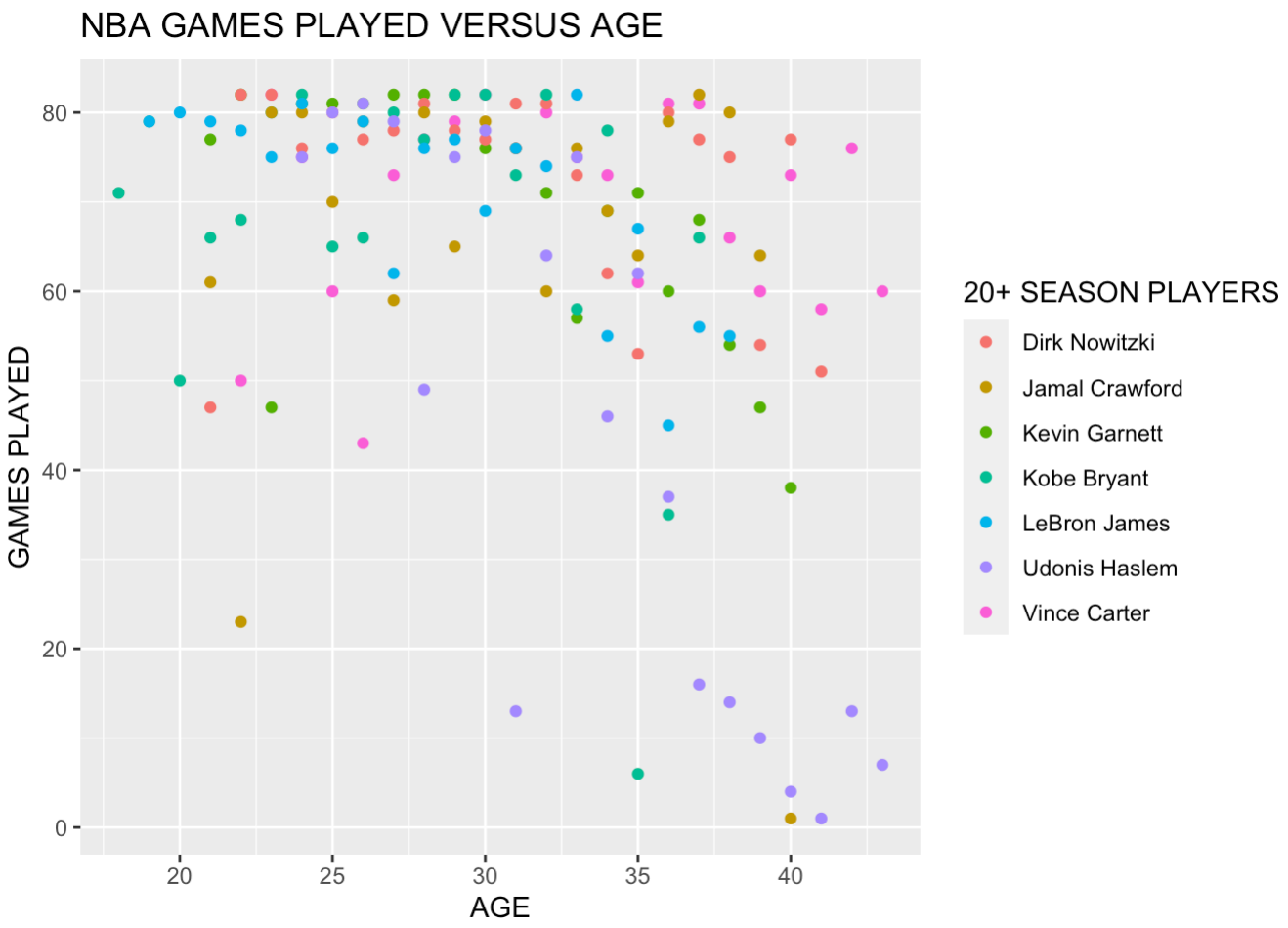
```
lebron5 <- nbaData %>%
  filter(PLAYER_NAME %in% c("LeBron James", "Kevin Garnett", "Vince Carter", "Kobe Bryant", "Udonis Haslem", "Dir
k Nowitzki", "Jamal Crawford")) %>%
  select(AGE,PTS, PLAYER_NAME)


ggplot(lebron5, aes(AGE,PTS, color = PLAYER_NAME)) +
  geom_point() +
  ggtitle("20+ SEASON PLAYERS POINTS PER GAME VERSUS AGE") +
  xlab("AGE") +
  ylab("POINTS PER GAME") +
  labs(color = "Players")
```



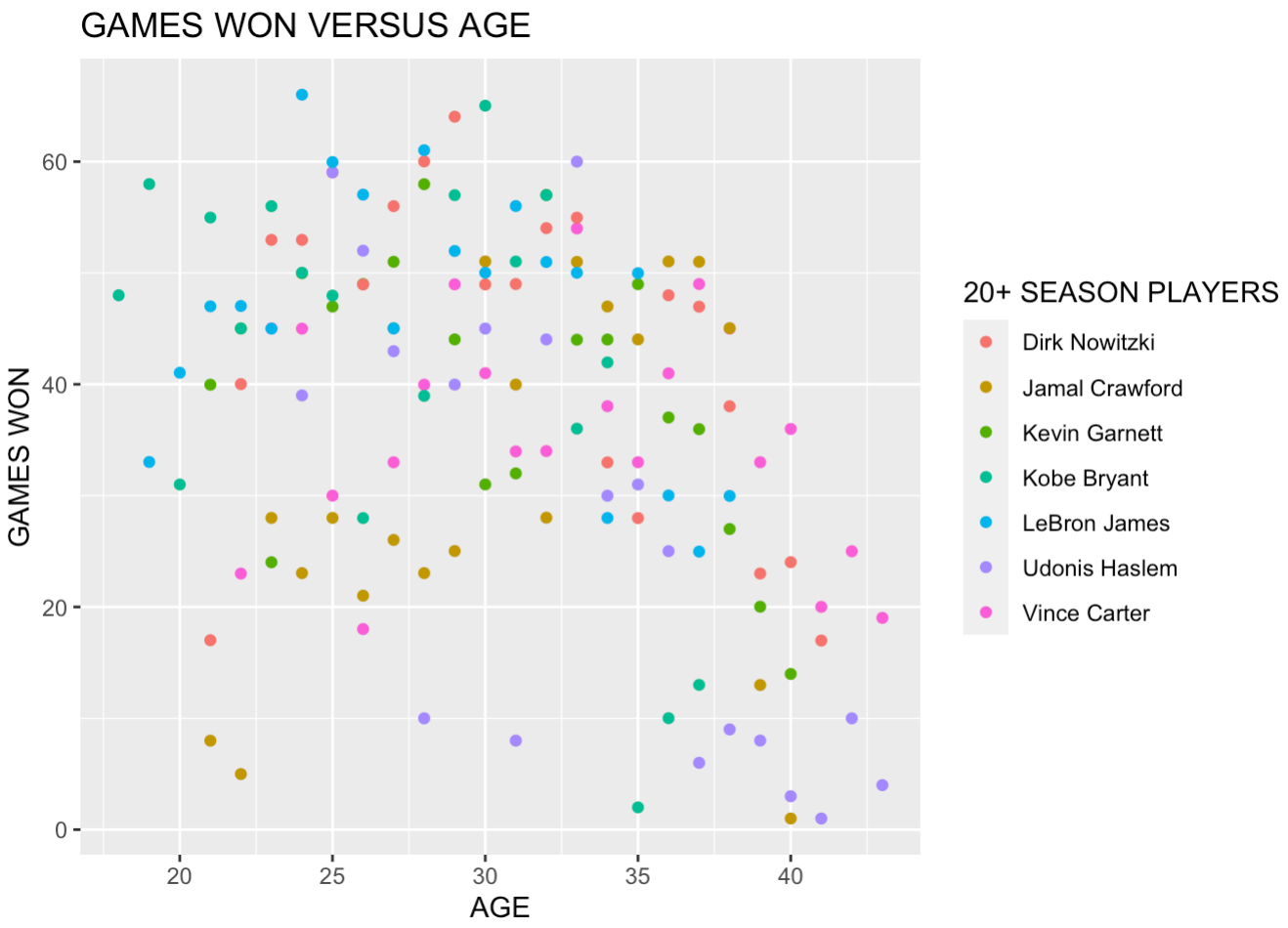
```
lebron5 <- nbaData %>%
  filter(PLAYER_NAME %in% c("LeBron James", "Kevin Garnett", "Vince Carter", "Kobe Bryant", "Udonis Haslem", "Dir
k Nowitzki", "Jamal Crawford")) %>%
  select(AGE,MIN, PLAYER_NAME)


ggplot(lebron5, aes(AGE,MIN, color = PLAYER_NAME)) +
  geom_point() +
  ggtitle("MINUTES PLAYED VERSUS AGE") +
  xlab("AGE") +
  ylab("MINUTES PLAYED PER GAME") +
  labs(color = "20+ SEASON PLAYERS")
```



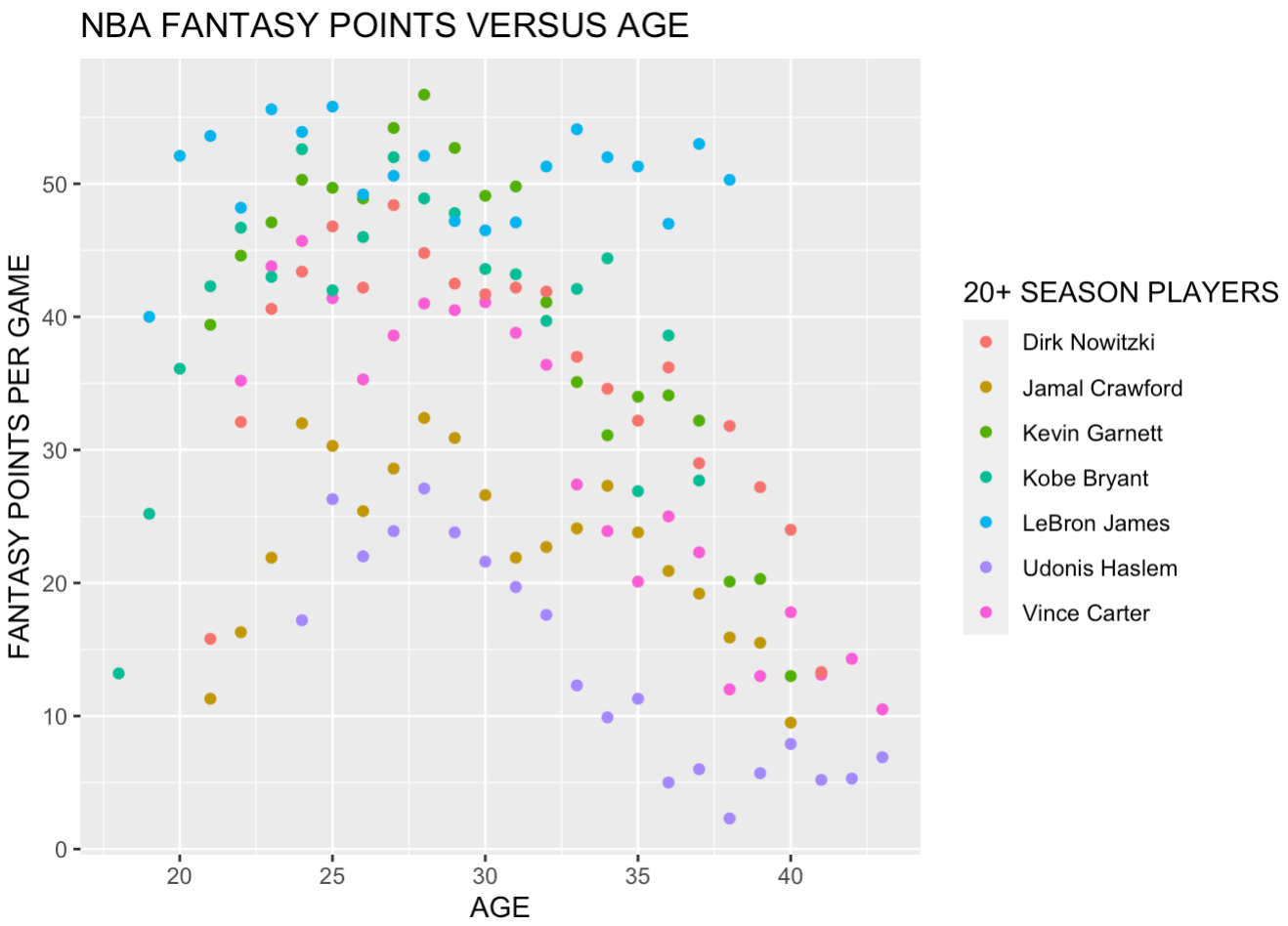
```
lebron5 <- nbaData %>%
  filter(PLAYER_NAME %in% c("LeBron James", "Kevin Garnett", "Vince Carter", "Kobe Bryant", "Udonis Haslem", "Dir
k Nowitzki", "Jamal Crawford")) %>%
  select(AGE,GP, PLAYER_NAME)


ggplot(lebron5, aes(AGE,GP, color = PLAYER_NAME)) +
  geom_point() +
  ggtitle("NBA GAMES PLAYED VERSUS AGE") +
  xlab("AGE") +
  ylab("GAMES PLAYED") +
  labs(color = "20+ SEASON PLAYERS")
```



```
lebron5 <- nbaData %>%
  filter(PLAYER_NAME %in% c("LeBron James", "Kevin Garnett", "Vince Carter", "Kobe Bryant", "Udonis Haslem", "Dir
k Nowitzki", "Jamal Crawford")) %>%
  select(AGE,W_PCT,GP, PLAYER_NAME)


ggplot(lebron5, aes(AGE,W_PCT*GP, color = PLAYER_NAME)) +
  geom_point() +
  ggtitle("GAMES WON VERSUS AGE") +
  xlab("AGE") +
  ylab("GAMES WON") +
  labs(color = "20+ SEASON PLAYERS")
```

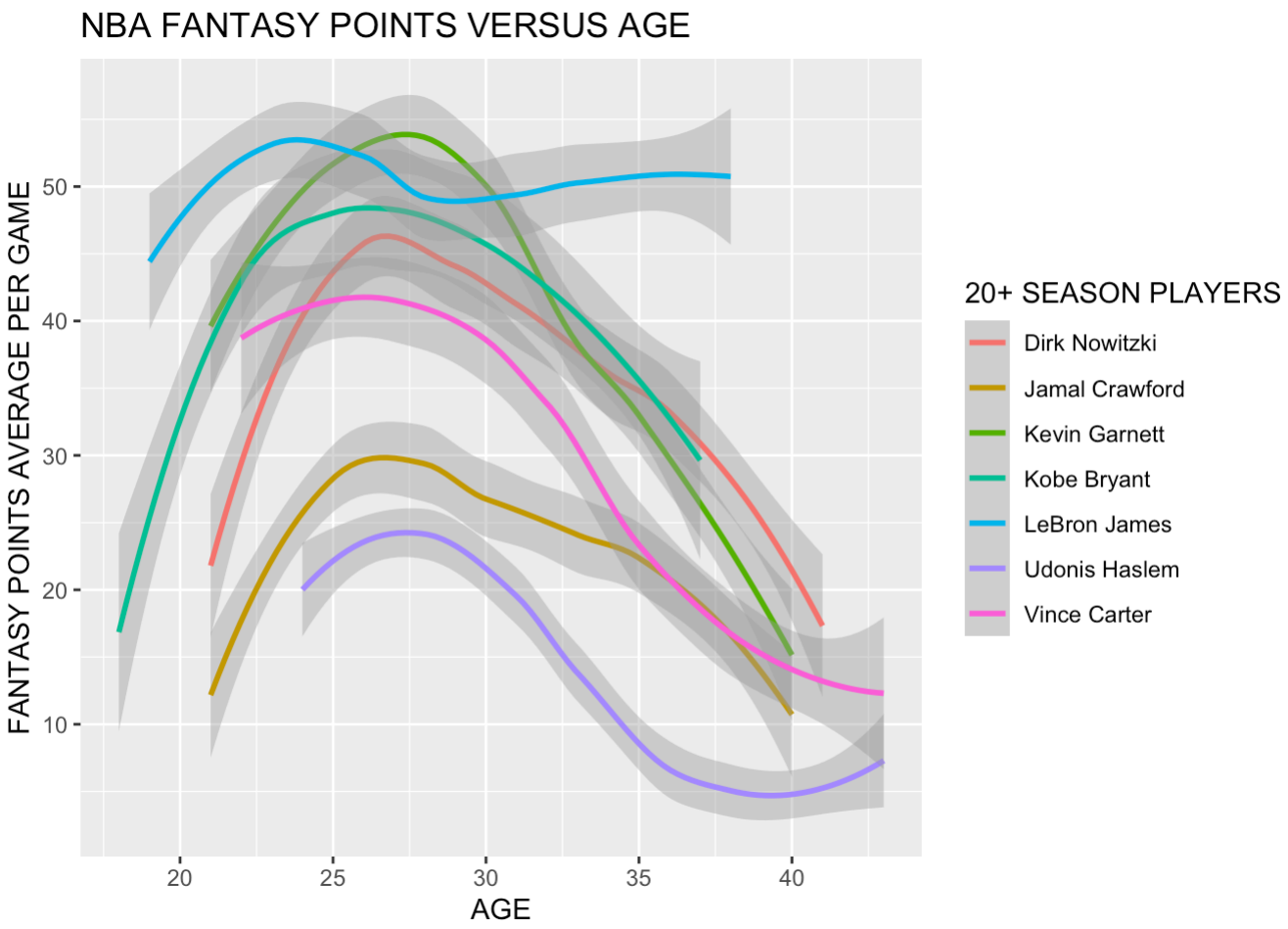


```
lebron5 <- nbaData %>%
  filter(PLAYER_NAME %in% c("LeBron James", "Kevin Garnett", "Vince Carter", "Kobe Bryant", "Udonis Haslem", "Dir
k Nowitzki", "Jamal Crawford")) %>%
  select(AGE,W_PCT,NBA_FANTASY_PTS, PLAYER_NAME)


ggplot(lebron5, aes(AGE,NBA_FANTASY_PTS, color = PLAYER_NAME)) +
  geom_point() +
  ggtitle("NBA FANTASY POINTS VERSUS AGE") +
  xlab("AGE") +
  ylab("FANTASY POINTS PER GAME") +
  labs(color = "20+ SEASON PLAYERS")
```

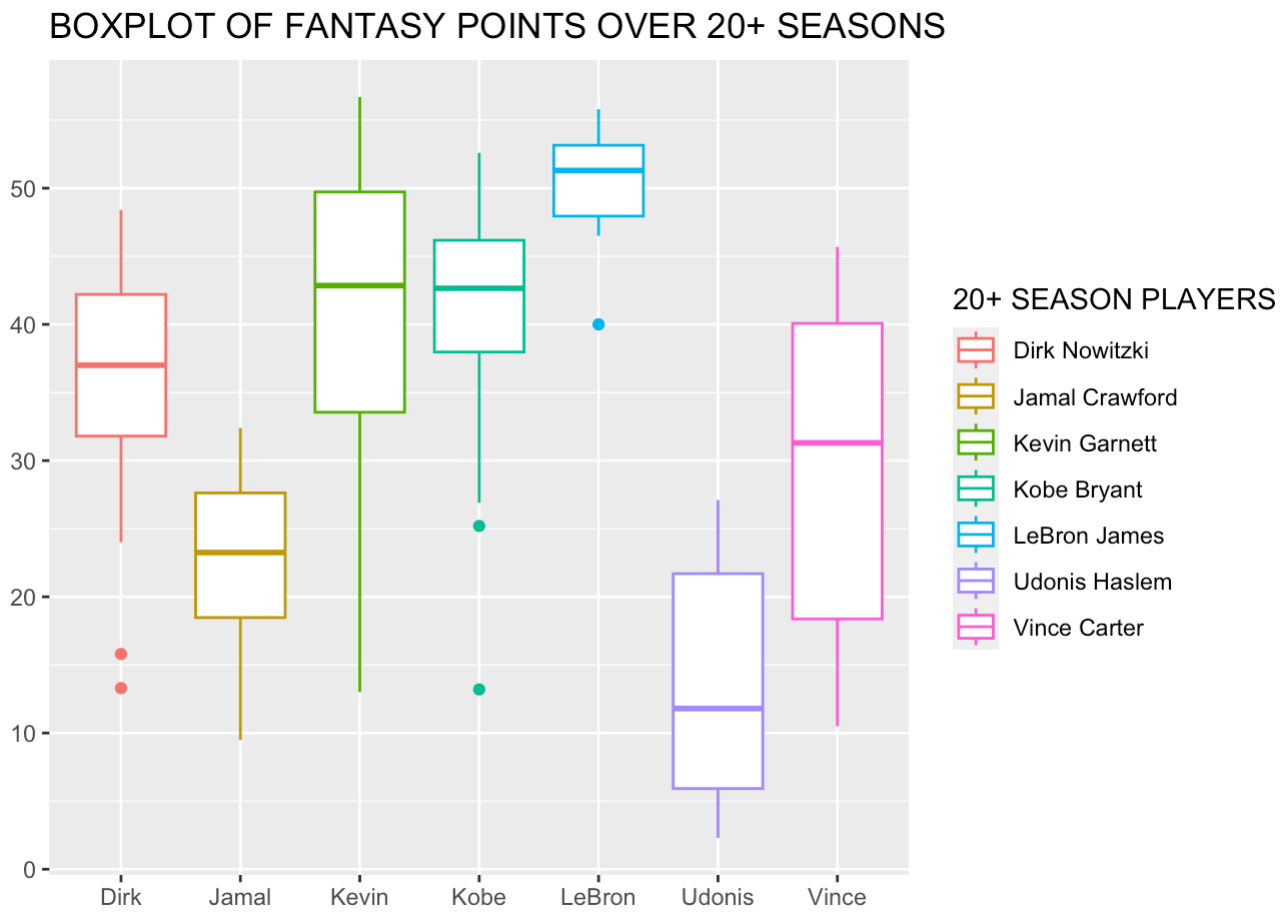


```
lebron5 <- nbaData %>%
  filter(PLAYER_NAME %in% c("LeBron James", "Kevin Garnett", "Vince Carter", "Kobe Bryant", "Udonis Haslem", "Dir
k Nowitzki", "Jamal Crawford")) %>%
  select(AGE,W_PCT,NBA_FANTASY_PTS, PLAYER_NAME)


ggplot(lebron5, aes(AGE,NBA_FANTASY_PTS, color = PLAYER_NAME)) +
  geom_smooth() +
  ggtitle("NBA FANTASY POINTS VERSUS AGE") +
  xlab("AGE") +
  ylab("FANTASY POINTS AVERAGE PER GAME") +
  labs(color = "20+ SEASON PLAYERS")
```

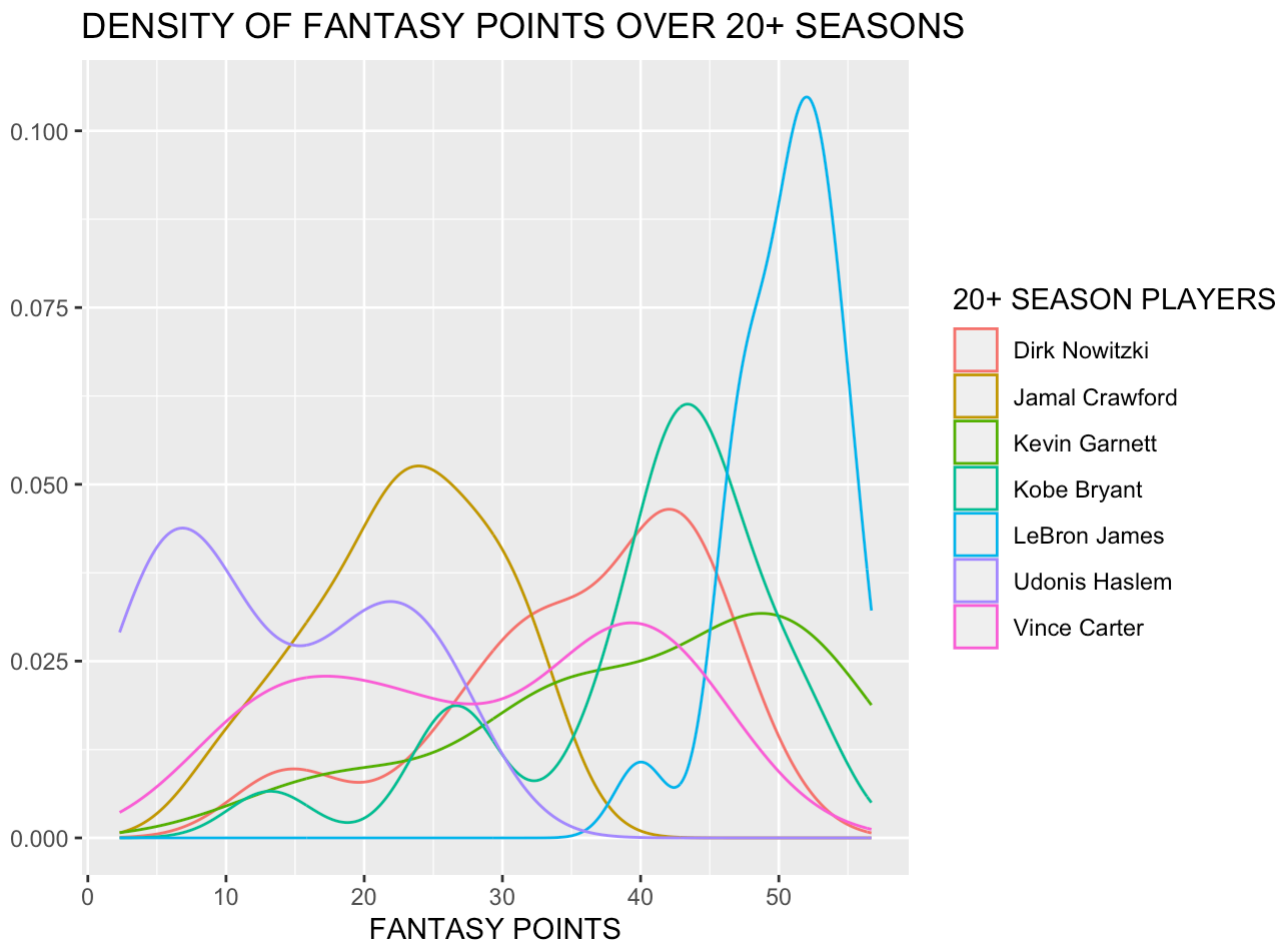
```
## `geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```




```
lebron5 <- nbaData %>%
  filter(PLAYER_NAME %in% c("LeBron James", "Kevin Garnett", "Vince Carter", "Kobe Bryant", "Udonis Haslem", "Dir
k Nowitzki", "Jamal Crawford")) %>%
  select(AGE,W_PCT,NBA_FANTASY_PTS, PLAYER_NAME,NICKNAME)


ggplot(lebron5, aes(NICKNAME,NBA_FANTASY_PTS, color = PLAYER_NAME)) +
  geom_boxplot() +
  ggtitle("BOXPLOT OF FANTASY POINTS OVER 20+ SEASONS") +
  xlab("") +
  ylab("") +
  labs(color = "20+ SEASON PLAYERS")
```




```
lebron5 <- nbaData %>%
  filter(PLAYER_NAME %in% c("LeBron James", "Kevin Garnett", "Vince Carter", "Kobe Bryant", "Udonis Haslem", "Dir
k Nowitzki", "Jamal Crawford")) %>%
  select(AGE,W_PCT,NBA_FANTASY_PTS, PLAYER_NAME,NICKNAME,GP,MIN,PTS,FGA,FGM,FG_PCT,OREB,DREB,AST,TOV,BLK,STL)


ggplot(lebron5, aes(NBA_FANTASY_PTS, color = PLAYER_NAME)) +
  geom_density() +
  ggtitle("DENSITY OF FANTASY POINTS OVER 20+ SEASONS") +
  xlab("FANTASY POINTS") +
  ylab("") +
  labs(color = "20+ SEASON PLAYERS")
```




```
cor.test(nbaData$AGE, nbaData$NBA_FANTASY_PTS)
```

```
##
##  Pearson's product-moment correlation
##
## data:  nbaData$AGE and nbaData$NBA_FANTASY_PTS
## t = 4.5739, df = 12844, p-value = 4.832e-06
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  0.02304888 0.05757875
## sample estimates:
##        cor
## 0.04032586
```

```
cor.test(lebron5$AGE, lebron5$NBA_FANTASY_PTS)
```

```
##
##  Pearson's product-moment correlation
##
## data:  lebron5$AGE and lebron5$NBA_FANTASY_PTS
## t = -6.8671, df = 141, p-value = 1.926e-10
## alternative hypothesis: true correlation is not equal to 0
## 95 percent confidence interval:
##  -0.6142948 -0.3666053
## sample estimates:
##        cor
## -0.5006273
```

```
linear <- lm(AGE~NBA_FANTASY_PTS,lebron5)
summary(linear)
```

```
##
## Call:
## lm(formula = AGE ~ NBA_FANTASY_PTS, data = lebron5)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -16.8590  -3.3276   0.9084   3.9875  11.0744
##
## Coefficients:
##                 Estimate Std. Error t value Pr(>|t|)
## (Intercept)     37.68172    1.12602  33.465  < 2e-16 ***
## NBA_FANTASY_PTS -0.21384    0.03114  -6.867 1.93e-10 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 5.413 on 141 degrees of freedom
## Multiple R-squared:  0.2506, Adjusted R-squared:  0.2453
## F-statistic: 47.16 on 1 and 141 DF,  p-value: 1.926e-10
```

```
linear2 <- lm(AGE~DREB+OREB+MIN,lebron5)
summary(linear2)
```

```
##
## Call:
## lm(formula = AGE ~ DREB + OREB + MIN, data = lebron5)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -16.411  -1.919   1.007   2.609   6.932
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) 42.23976    1.22887  34.373  < 2e-16 ***
## DREB         1.25105    0.22311   5.607 1.07e-07 ***
## OREB        -3.96956    0.65211  -6.087 1.06e-08 ***
## MIN         -0.42265    0.04905  -8.616 1.37e-14 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.313 on 139 degrees of freedom
## Multiple R-squared:  0.5309, Adjusted R-squared:  0.5208
## F-statistic: 52.44 on 3 and 139 DF,  p-value: < 2.2e-16
```

```
linear3 <- lm(NBA_FANTASY_PTS~MIN+FGA,lebron5)
summary(linear3)
```

```
##
## Call:
## lm(formula = NBA_FANTASY_PTS ~ MIN + FGA, data = lebron5)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -10.3903  -3.1177  -0.4992   2.9280  12.0484
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -4.1343     1.4940  -2.767  0.00642 **
## MIN           0.5429     0.1009   5.381 3.04e-07 ***
## FGA           1.4520     0.1583   9.173 5.36e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 4.81 on 140 degrees of freedom
## Multiple R-squared:  0.8928, Adjusted R-squared:  0.8913
## F-statistic: 582.9 on 2 and 140 DF,  p-value: < 2.2e-16
```

```
cor(lebron5$NBA_FANTASY_PTS,lebron5$MIN)
```

```
## [1] 0.9101376
```

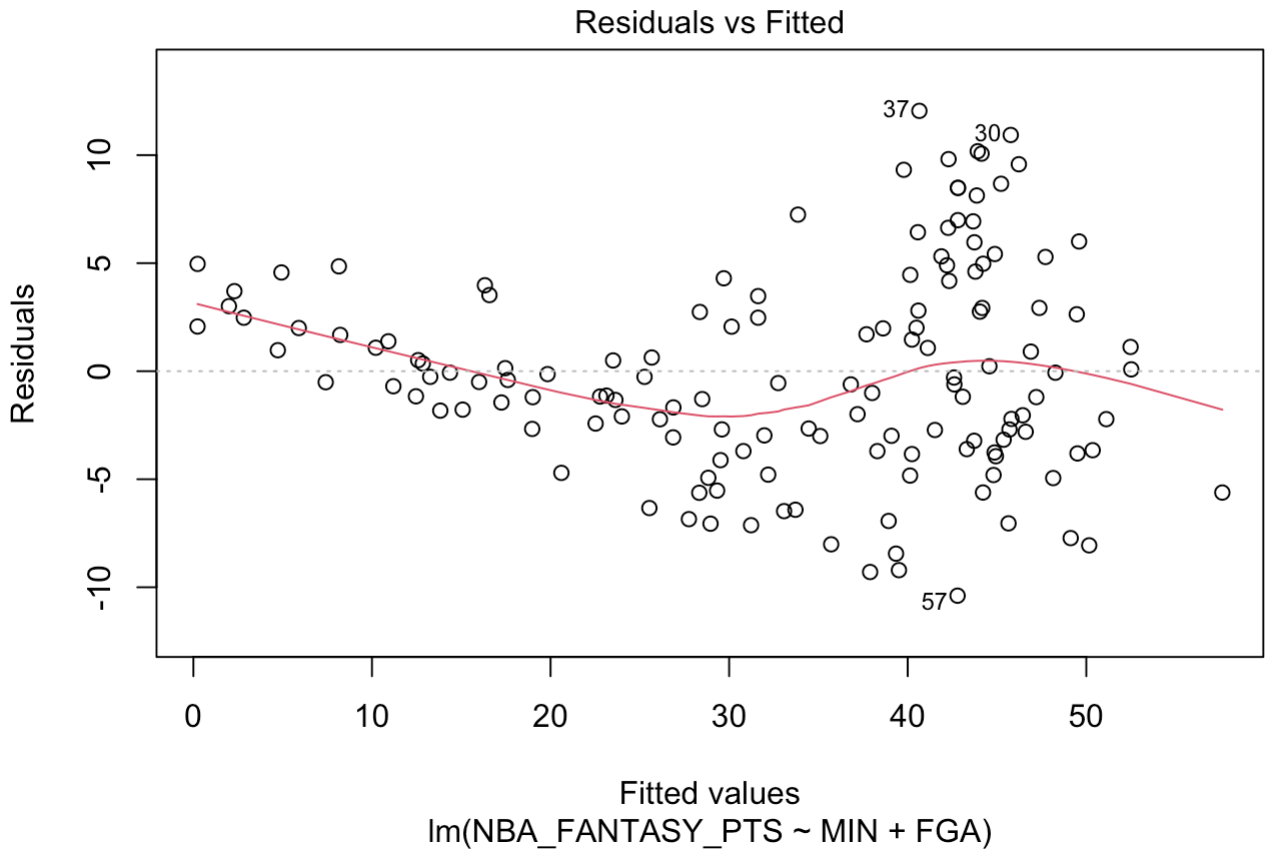
```
cor(lebron5$NBA_FANTASY_PTS,lebron5$FGA)
```

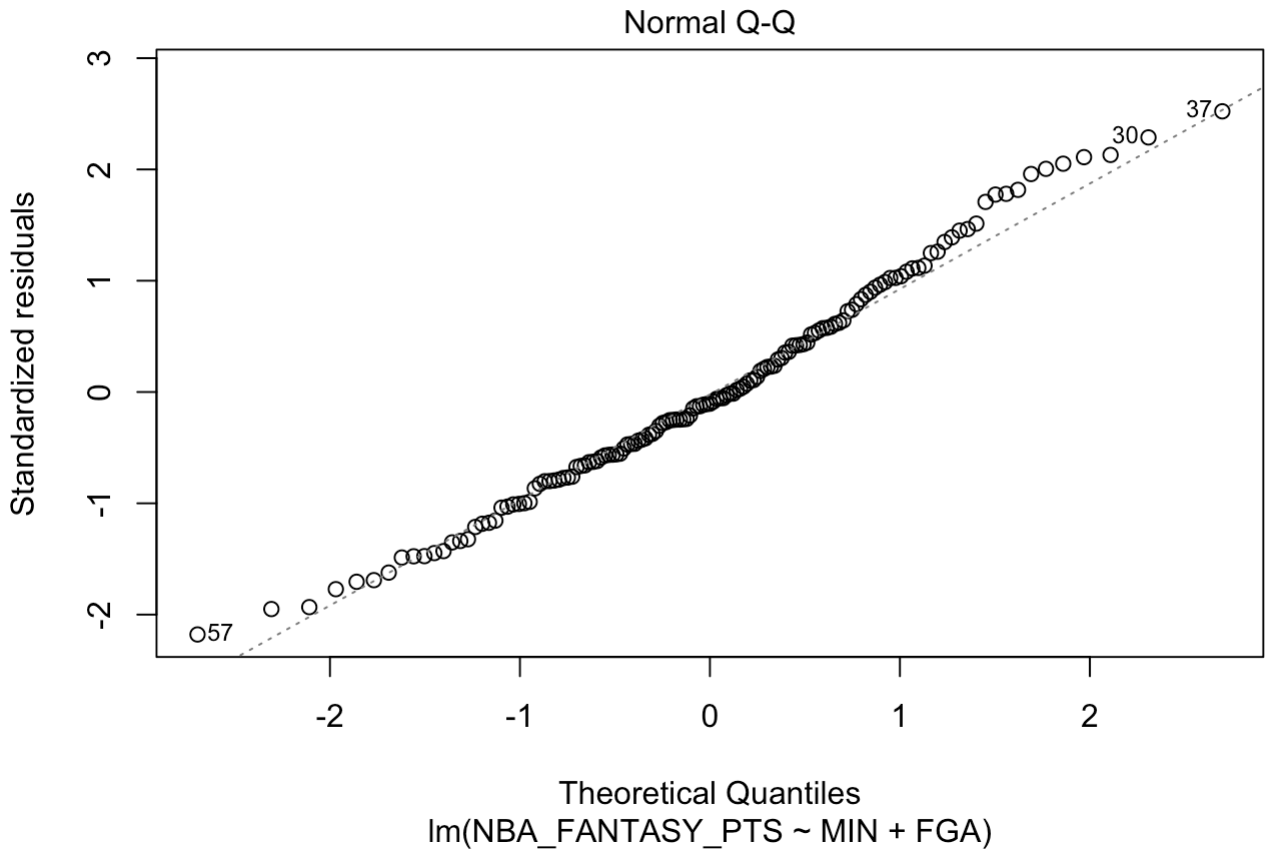
```
## [1] 0.9330645
```

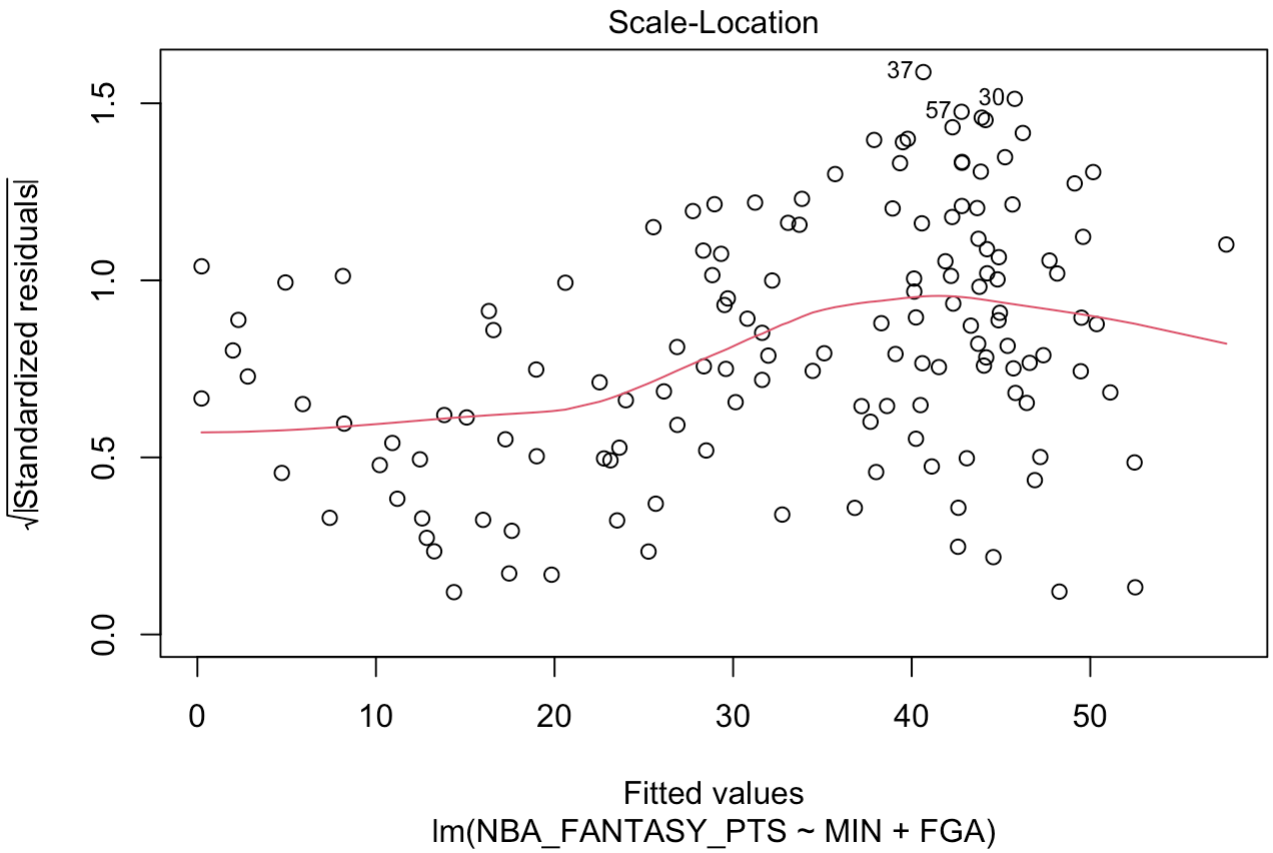
```
cor(lebron5$FGA,lebron5$MIN)
```

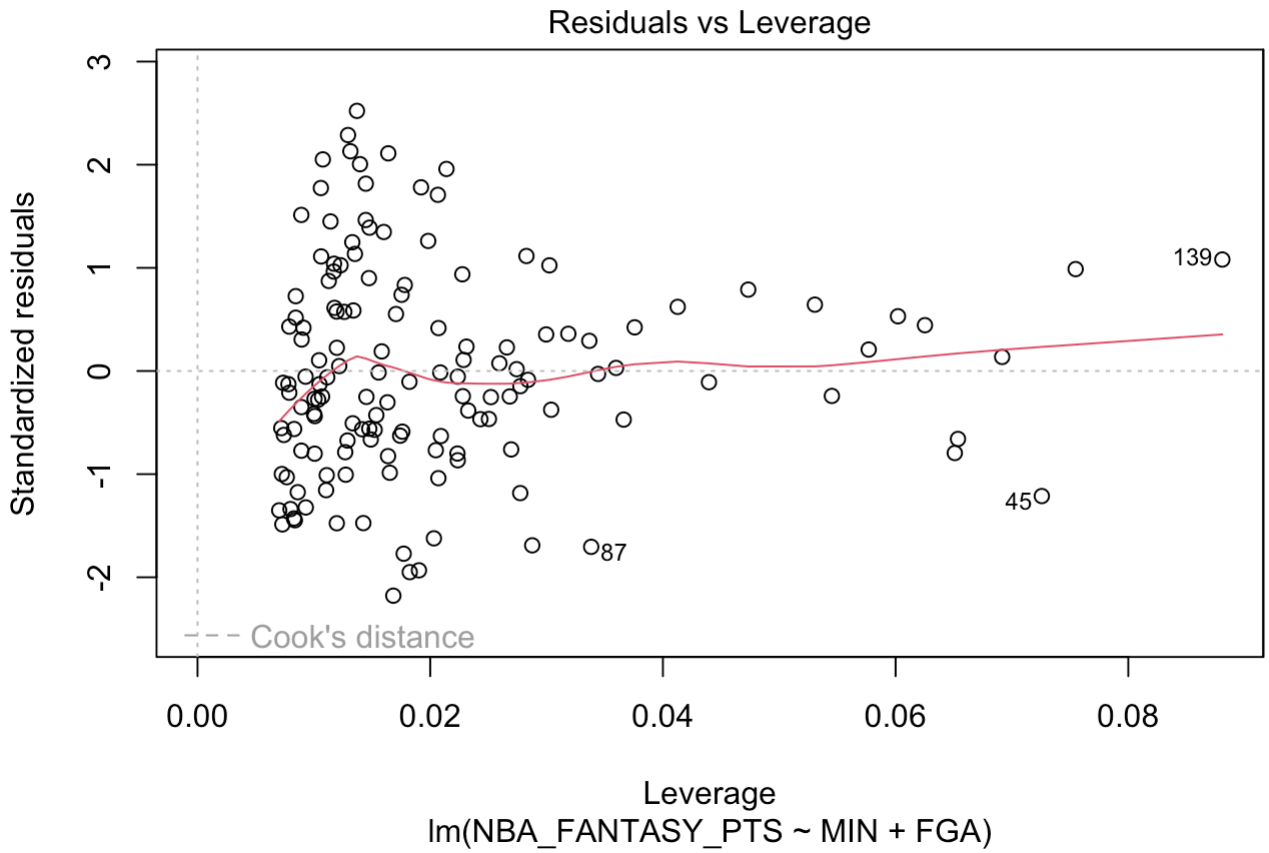
```
## [1] 0.9088611
```

```
plot(linear3)
```

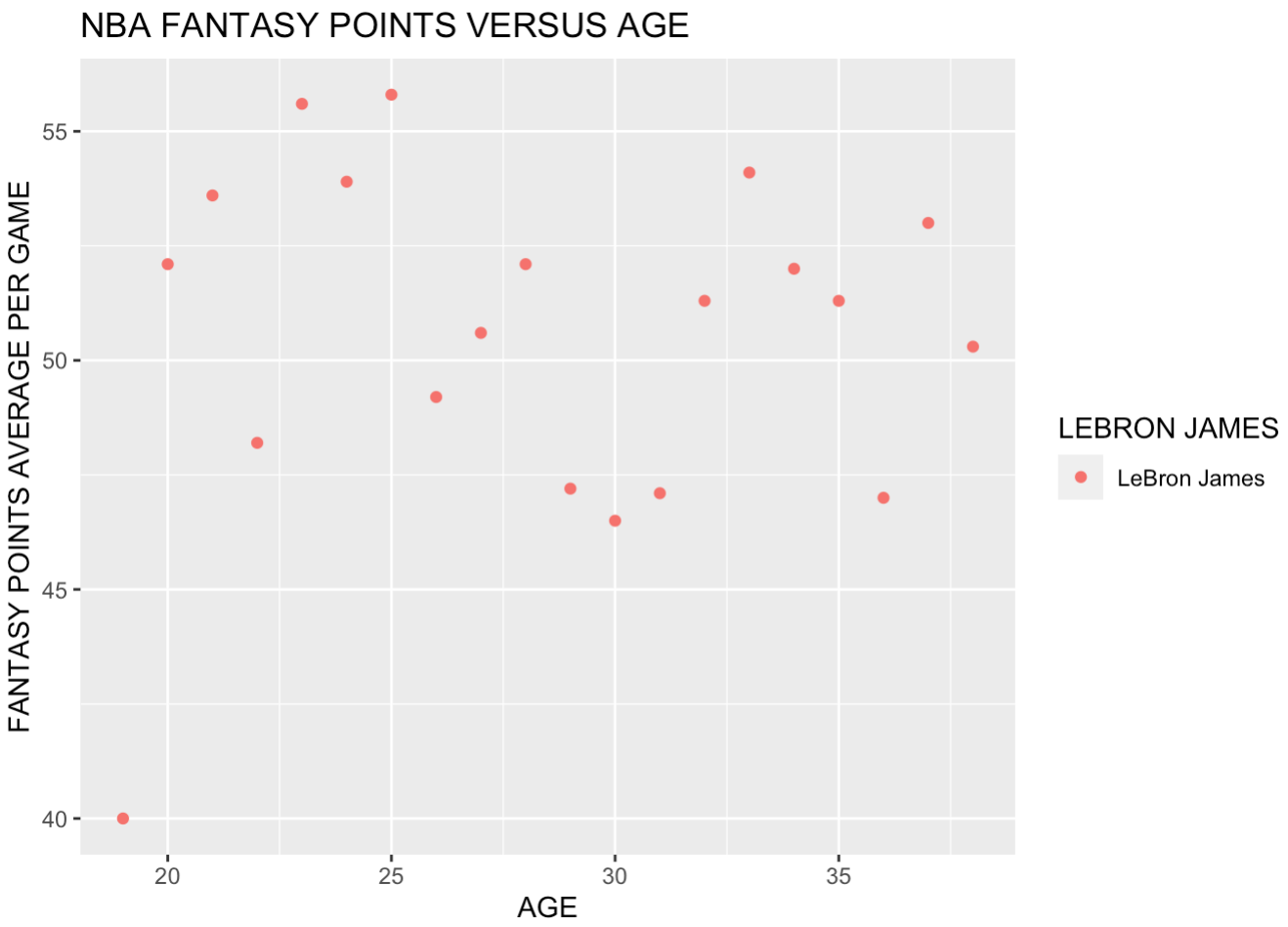

Residuals vs Fitted
Residuals
Fitted values
lm(NBA_FANTASY_PTS ~ MIN + FGA)
37
30
57
10
5
0
-5
-10
0
10
20
30
40
50


Normal Q-Q
Standardized residuals
Theoretical Quantiles
lm(NBA_FANTASY_PTS ~ MIN + FGA)
3
2
1
0
-1
-2
-2
-1
0
1
2
37
30
57


Scale-Location
37
30
57
√|Standardized residuals|
0.0
0.5
1.0
1.5
0
10
20
30
40
50
Fitted values
lm(NBA_FANTASY_PTS ~ MIN + FGA)




```
lebron5 <- nbaData %>%
  filter(PLAYER_NAME %in% c("LeBron James")) %>%
  select(AGE,W_PCT,NBA_FANTASY_PTS, PLAYER_NAME,MIN,FGA)


ggplot(lebron5, aes(AGE,NBA_FANTASY_PTS, color = PLAYER_NAME)) +
  geom_point() +
  ggtitle("NBA FANTASY POINTS VERSUS AGE") +
  xlab("AGE") +
  ylab("FANTASY POINTS AVERAGE PER GAME") +
  labs(color = "LEBRON JAMES")
```




```
linear3 <- lm(NBA_FANTASY_PTS~MIN+FGA+AGE+W_PCT,lebron5)
summary(linear3)
```

```
##
## Call:
## lm(formula = NBA_FANTASY_PTS ~ MIN + FGA + AGE + W_PCT, data = lebron5)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -3.5283 -1.6905 -0.1454  1.7596  4.5583
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept) -51.8357    27.6251  -1.876  0.08019 .
## MIN           1.2284     0.6304   1.949  0.07029 .
## FGA           1.3528     0.5270   2.567  0.02146 *
## AGE           0.5311     0.2364   2.247  0.04010 *
## W_PCT        21.7213     5.9902   3.626  0.00249 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.618 on 15 degrees of freedom
## Multiple R-squared:  0.6207, Adjusted R-squared:  0.5196
## F-statistic: 6.137 on 4 and 15 DF,  p-value: 0.003932
```